Designing a Multimodal AI Debate Partner

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Use Case and Goals

Research shows that debating enhances critical thinking, writing quality, academic performance, and confidence [1]. However, debating clubs are often restricted to privileged schools, excluding marginalised students from its benefits [2]. Furthermore, learners often struggle with speaking anxiety, structuring arguments, and handling counterarguments [3]. AI could reduce these barriers by providing accessible and unlimited debate practice. Nevertheless, poor design may reinforce bias, provide misinformation, hinder critical thinking, and decrease confidence.

This project explores these design implications through designing and prototyping an AI system called "DebateAble", that aims to improve debating and argumentative skills through multimodal simulation. The prototype is evaluated through a small mixed methods user study, examining how interaction modality and learning approach influence user experience, alongside associated value trade-offs and ethical concerns.

The primary user group is university students due to their engagement in debating, research defense, and argumentative writing. Secondary stakeholders include lectures, debate societies, and debate coaches. Tertiary stakeholders comprise of AI developers and University IT services.

System Design and Implementation

The project began with brainstorming a list of potential use cases. Adopting a Human-Centered AI design approach enabled refinement of concepts based on students needs [4]. Elements of participatory co-design [5] were incorporated as four shortlisted ideas were presented to ten Cambridge students, who contributed design ideas and contextual considerations (see Appendix E). A debating system was the most preferred use case, as students believed it could improve oral and written argumentation.

Based on these insights, a hand sketched prototype was developed (see Appendix C). The system has two interaction modalities: text and audio. Following best-practice recommendations for co-designing conversational agents [6], the low-fidelity prototype was presented to three students to elicit user values and inform revisions.

Using a value-sensitive design framework [7, 8], the following values were identified:

- **Autonomy:** Users want to develop critical thinking rather than use AI generated answers
- **Control:** Adjustable levels of aggression to prevent user discouragement
- **Universal Usability:** Support users with varying skill levels
- **Transparency:** Evidence based response to ensure trust and fairness

I translated these values [9] into a three-level scaffolded learning structure:

- **Level 1 (Guided):** Supports novices through encouraging language, non-confrontational rebuttal, and socratic questioning to promote critical thinking instead of prescribing arguments.
- **Level 2 (Sparring):** Directly debate AI with real-time feedback. Difficulty is adjusted dynamically based on a user's argument, inspired by adaptive tutoring designs [10].
- **Level 3 (Advanced):** Debate an aggressive but respectful AI opponent. Feedback is given only post-debate.

User and AI persona's were developed to inform this scaffolding approach and system prompt design (see Appendix F).

This scaffolding structure aligns with Vygotsky's Zone of Proximal Development [11], which emphasises gradual assistance based on a learner's competence. It also supports autonomy and competence needs from self-determination theory [12].

Avatars were includes to increase engagement and interaction [13, 14]. Avatars were all abstract to minimize bias and avoid reinforcing stereotypes [15].

The prototype was initially developed in Figma and iteratively refined to improve visual fidelity using heuristic evaluation [16].

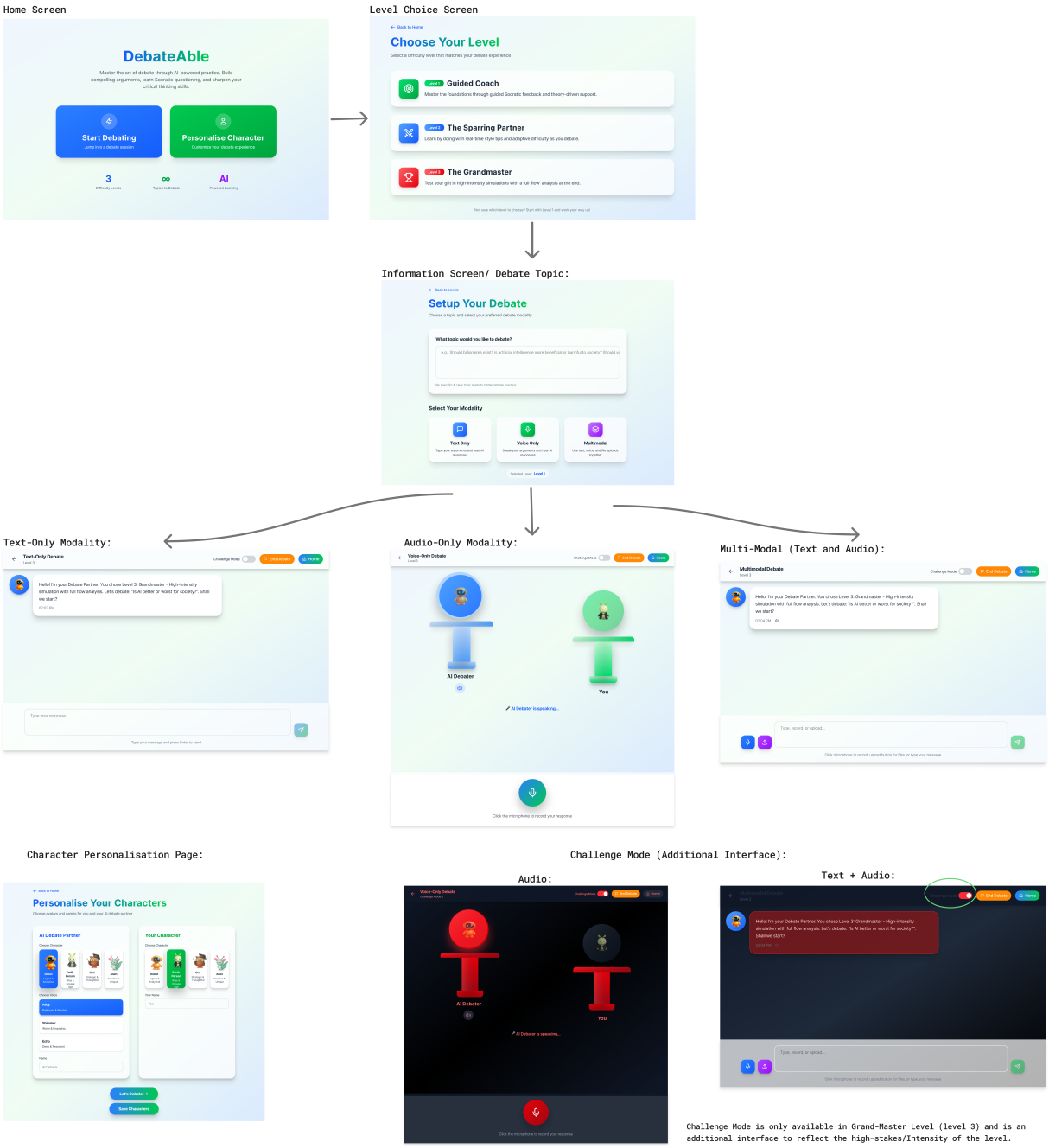

Figure 1: Finalised prototype interface showing various screens. Challenge mode is an optional interface available only in level 3. (See Appendix C for earlier versions).

Methods: RQS, Evaluation Methods & Metrics

Research Questions:

- **RQ1:** How does interaction modality (text-only, audio-only, multimodal) influence argument quality, perceived preparedness and learning, and modality preference in AI-supported debate practice?

- **RQ2:** How does scaffolded progression versus a single advanced level affect user experi-

ence, perceived learning, and cognitive demand?

- **RQ3:** What unintended risks and ethical concerns emerge when interacting with an AI debate partner?

Evaluation Methods

Nine Cambridge students (5 males, 4 females; ages 21–24) voluntarily participated. Modality was altered between participants; text only, audio only, or multimodal (user could input text or speech and received both text and spoken AI responses). The fourth configuration tested scaffolding by comparing Levels 1-2 with Level 3 using a counterbalanced within-subject approach.

Each session followed a structured protocol (see Appendix D):

1. Participants spent 8 minutes writing an augmentative paragraph on a chosen topic
2. Interacted with different system levels while providing think-aloud feedback
3. Completed survey and interview questions before revising their original paragraph

Each session lasted 1.5-2 hours, leading to a small sample size (n=9) and motivating a reflexive thematic analysis [17]. Argument quality was assessed via blind grading by both a human evaluator and an LLM-as-a-Judge *(Gemini-3.0-flash)* using a rubric based on Toulmin's model [18] (found in python notebook).

Survey questions (See Appendix A.3) were adapted from Intrsintic Motivation Inventory (enjoyment) [19], NASA-TLX (cognitive load) [20], and SUS (usability) [21] inspired questionnaires. Appendix B contains qualitative interview questions.

Results

RQ1 (Modality): Due to small sample size, no definitive claims can be made regarding modality's effect on writing quality. However, exploratory analysis reveals improvement in paragraph quality after using the debater system across modalities, with largest gains observed for multimodal users (see Table 1), followed by audio-only, and smallest for text-only. Similarly, multimodal users reported higher perceived learning gains.

Modality	Human			LLM		
	Pre	Post	Δ	Pre	Post	Δ
Text	19.3	22.0	2.7	18.0	20.3	2.3
Audio	18.7	22.0	3.3	21.3	24.0	2.7
Multimodal	16.7	20.7	4.0	18.7	23.3	4.6
Overall	18.2	21.6	3.4	19.3	22.6	3.3

Table 1: Measurements of Argument quality. Pre is paragraph scores before using AI, Post is after using AI.

Modality did not affect preparedness for writing versus oral argumentative tasks, with users feeling the app prepared them better for both (see Table 2). However, audio and multimodal users reported higher modality preference and naturalness compared to text only users. Interestingly, 2/3 multimodal users preferred text input due to discomfort with audio recordings. Additionally, though the social presence of AI was low overall, it was significantly higher in audio and multimodal groups.

Measure	Text	Audio	Multimodal
Essay preparedness (Q1)	4.00	4.33	4.33
Debate preparedness (Q2)	4.00	4.33	4.33
Skills gained (Q3–Q4)	3.83	4.00	4.50
Usability (Q5–Q6)	4.17	4.33	4.17
Modality naturalness & preference (Q7–Q8)	3.50	4.00	4.33
Social presence (Q9)	1.00	1.67	1.67

Table 2: Mean survey responses by modality (n = 3 per condition, scale 1–5, questions in Appendix A.3)

RQ2 (Levels 1-2 vs Level 3): Participants had higher enjoyment and perceived ability (see figure 2) in Levels 1-2, whereas, Level 3 elicited higher pressure and cognitive effort as users felt they had to work harder to keep up with the AI.

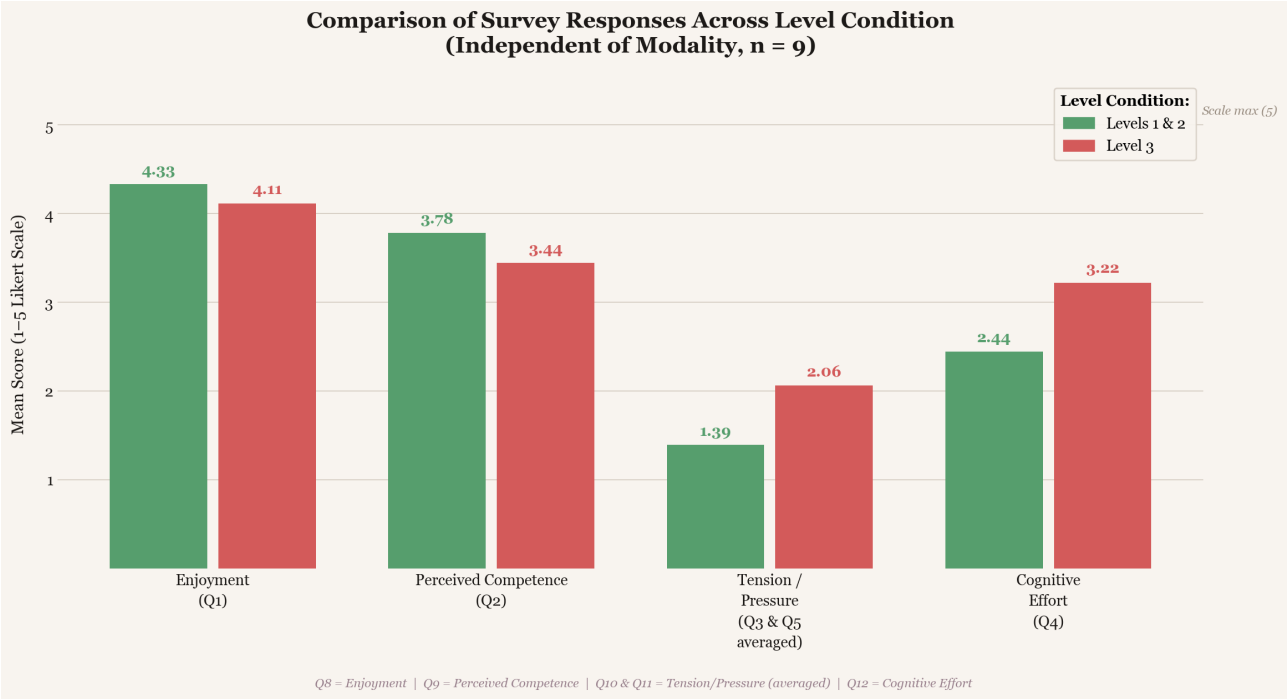

Figure 2: Bar chart demonstrating mean survey rating by level condition. (full questions in Appendix A.2).

Nevertheless, 7/9 users preferred the advanced level, stating that although it was sometimes frustrating, it encouraged more engagement and enhanced their learning. The other two users preferred Level 2, claiming it allowed more free reflective thinking and they could move to Level 3 when they felt more ready. Despite this, no user wanted one level exclusively, noting that earlier levels are more useful when they are less familiar with a topic.

RQ3 (Risks): Participants reported low emotional attachment to the system however, majority agreed that richer anthropomorphic avatars (video based) could increase emotional reliance particularly for vulnerable users.

All participants reported that interacting with the AI debate system changed how they approached forming arguments or debating. Most changes were positive such as exposure to counter-arguments however, few noted a reconsideration of their own positions, raising concerns about subtle persuasion risk and belief off-loading.

Automation bias was evident as 7/9 participants reported high trust in the AI's output despite acknowledging limited verification mechanisms.

Privacy concerns were raised by 6/9 participants regarding the use of voice data and debate topics (could reveal sensitive beliefs).

Discussion

Results reveal the following value trade-offs:

- **Learning Effectiveness vs User Autonomy:** Level 3 increased perceived learning but reduced comfort and control.

- **Critical Thinking vs Persuasion:** Realistic debates promoted critical thinking through diverse counter-arguments, but strong persuasive influence could inadvertently encourage belief offloading, especially in vulnerable users.
- **Immersiveness vs Privacy:** Including audio modality (users or AI speaking) enhanced immersion and social presence. However, not all users were comfortable sending audio, which raises greater privacy concerns than text.

Analysis suggests multimodality is the optimal modality for this system as it supports user autonomy and control [22] by allowing optional audio input. Additionally, it aligns with ability based design [23] by improving accessibility to users with diverse abilities. Consistent with HCI literature [24], multimodality can also improve engagement and learning.

The trade-off between learning efficiency and user autonomy supports a scaffolding approach with three Levels, promoting user control by making difficulty adjustable rather than system imposed.

Unintended risk and Potential Mitigation:

- **Belief Offloading and Automation Bias:** Users opinions can be manipulated by AI [25]. Mitigation: reflective prompting, present multiple perspectives, and signal uncertainty to support trust.
- **Hallucinated or Bias Arguments:** AI may generate incorrect or bias content. Mitigation: citation of sources with links (retrieval based grounding).
- **Normative Framing:** System may reinforce dominant debating styles or decrease performance for non-native English speakers. Mitigation: configurable debate styles and accessibility testing.

Limitations, Future Work, and Conclusion

Small sample size from a high achieving academic population limits generalisability of findings. Using LLM-as-a-judge scoring may reward argument styles similar to data distribution rather than superior reasoning [26]. The study did not control for topic familiarity which likely confounded learning gains across conditions.

Future work can include larger sample sizes, multiple human expert raters, and longitudinal studies to see if learning gains persist over time. Participation design workshops with educators, debate clubs, or law programs can further improve system design.

In conclusion, DebateAble shows potential for developing argumentative skills, particularly when using multimodality and a scaffolded learning approach. However, designing such a system requires weighing value trade-offs and risks to ensure a human-centered approach that evolves with users needs.

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Appendix

A Survey and Interview Materials

I would like to thank all the participants for their time and thoughtful feedback.

A.1 Pre-Interaction Survey

Background
Using a 5-point Likert Scale (1 = Never, 5 = Very Often) answer the following:

- Have you used AI tools (e.g. ChatGPT) for essays or debate practice before?
- How often do you engage in:
  - Writing argumentative essays
  - Formal or informal debating (e.g., debating with friends or family)

  Using a 5-point Likert Scale (1 = Strongly Disagree, 5 = Strongly Agree) Please rate your agreement with the following:

  - I feel confident in my ability to debate.
  - I feel confident in my argumentative writing skills.
  - I feel stressed when I have to defend an argument.

A.2 Post-Level Survey (Completed After Each Condition)

Participants completed this survey after both Level 1–2 and Level 3 conditions (counterbalanced).

Intrinsic Motivation Inventory (IMI)
Using 5-point Likert Scale: 1 = Not at all true, 5 = Very true

1. Enjoyment: "I enjoyed doing this debate activity very much."
2. Competence: "I think I was pretty good at this activity."
3. Pressure: "I felt tense or pressured while doing this activity."

NASA Task Load Index (Adapted)
Using 5-point Likert Scale: 1 = Very low demand, 5 = Very high demand

4. Mental Demand/Effort: "How hard did you have to work to keep up with the AI?"
5. Frustration: "How annoyed, stressed, insecure, or discouraged did you feel during this task?"

Additional Item

- The AI felt most like a: Tutor / Opponent / Tool

A.3 Post-App Interaction Survey

Perceived Confidence and Preparedness
Using 5-point Likert Scale: 1 = Strongly Disagree, 5 = Strongly Agree

1. Using this app made me feel better prepared for an argumentative essay than before.
2. Using this app made me feel better prepared for a verbal debate than before.
3. I feel more confident debating this topic than before.
4. I can now see and argue both sides of the topic more clearly.

Using 5-point Likert Scale: Usability Scale (Adapted)
Scale: 1 = Strongly Disagree, 5 = Strongly Agree

5. I think I would use this app frequently for learning.
6. I think this app was easy to use.

Modality and Trust
Using 5-point Likert Scale: 1 = Strongly Disagree, 5 = Strongly Agree

7. The interaction modality (text/audio/multimodal) felt natural.
8. I would choose this modality over others for future debate practice.
9. The AI felt like a conversational partner rather than a tool.
10. I trusted the AI's arguments and feedback.

B Interview Protocol

B.1 After Experiencing Both Level Conditions

- Which learning paradigm did you prefer? Why?
- Which felt most realistic?
- In which condition did you feel you learned more?
- If you preferred a specific level, would you still want the option of other levels?

B.2 Post-Study Interview Questions

General Experience

- Can you describe your experience using the Debator app?
- Did the AI change how you approached debating or forming arguments?

Anthropomorphism and Social Relationship

- Did the AI feel human-like or more like a tool? Why?

- Could people form an emotional attachment to this system?
- How would increased personalisation (e.g., video or voice options) change your experience?

Influence and Manipulation

- Did the AI influence your opinions or arguments?
- Could this system influence vulnerable users?
- What made you trust or distrust the AI?
- Could users become dependent on this system?

Privacy

- Did you have any privacy concerns?
- If using voice, did you think about where your audio data might go?

Final Reflection

- Would you use this system for real learning? Why or why not?
- What is the single most important improvement you would make?

C Prototype Iterations

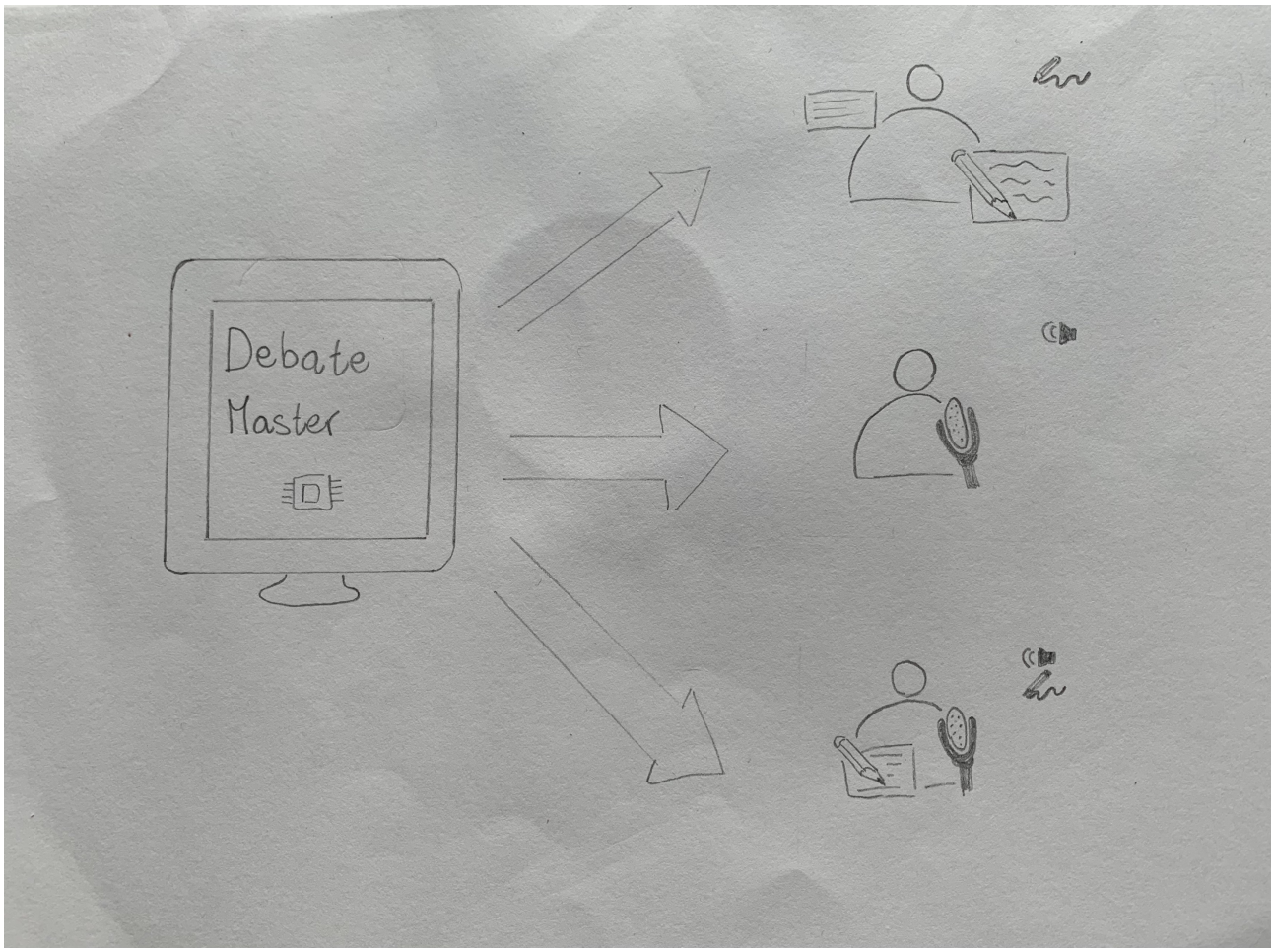

Figure 3: Abstract Sketch of prototype presented to 3 Cambridge students for feedback and idea generation.

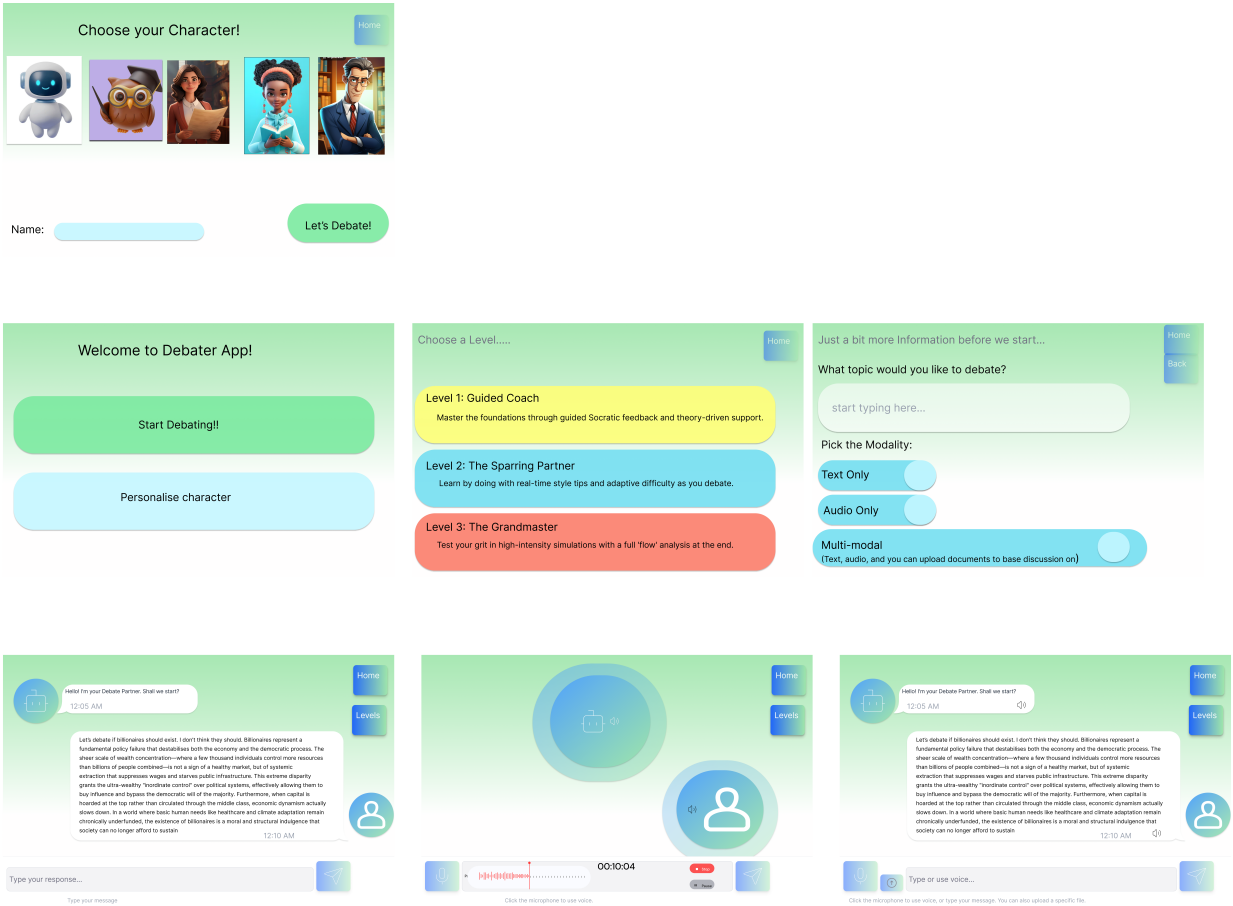

Figure 4: Initial Interface Design on Figma. Then Figma Make and feedback was used to inform and come up with the final design.

D User Study Protocol

Each session followed a structured protocol, here are the full details of how the user study was conducted:

1. Pre-study survey and interview questions to capture baseline experiences
2. Participants had 8 minutes to write an argumentative paragraph on a chosen topic (without external resources)
3. Participants had 10 minutes to test out each level. After testing a condition (scaffolded vs challenge) subjects filled out a survey. Think-aloud notes were recorded.
4. Post-study survey and interview questions
5. Participants had 8 minutes to revise or rewrite their original paragraph

E Brainstorming Multimodal AI Use Cases

Ideas	Comments
Journaling AI App: • **Would help prompt what to journal about** • **Would help you reflect on your feelings or what you wrote** • **Can give you analysis based on what your write** • **Can give a year, months, day summaries** **Can include a cartoon version at the end of the year of the entries/what you talked about (video wrap)**	• Concerns it would defeat the point of journaling; to get away from technology • Concerns of AI dependency/trust/ Replacement for counseling • Most students experience high stress periods or deal with mental issues, potential help but can't substitute counseling • People that don't journal would find it useful to start • Other's have found it useful but claim they use similar apps; Apple's built-in Journal app • LLM should not judge or directly claim, more questions and allow reflection • More difficult to study in short term; to really get a feel for users they would need to test it out longer amount of time • Difficult to test/ideally would need participants to take the app home and use it and then interview on experience • Longitudinal study would be ideal • Similar studies have been done, similar apps do exist
AI Debating App: • **Helps your learn the art of debating but debating with an AI through text or audio** • **Can input other texts of pictures for reference** • **Can also test you on specific questions (different mode)**	• Would want different levels of debating based on beginner to a person who knows the subject more • Would want to debate on controversial topics • Useful for oral exams or presentation practice • Some users found most useful • Other claim they wouldn't trust the evidence or couldn't accurately imitate human questions/debate concerns • Becomes easily annoying since AI can easily be always right (has so much more information that you)
Traveling App: • **Helps plan trips of budgets** • **Get ideas of where to go based on what you want to do or itinerary** • **Helps plan what to pack** **Help coordinate dates with friends**	• Some found very useful but would need to not overwhelm user and have everything in one app at the same time • Don't want AI to make decisions for them but rather make recommendations • Many other Apps already exist
International Student guidelines:	• need information to be reliable and up to date

Figure 5: List of Idea's and key notes taken from discussions with 10 Cambridge Students.(Page 1)

• **Helps international students transition to UK** • **Provide information on visa, healthcare system, opening bank accounts, different educational system, phone, ect.**	• Risk of hallucination is particularly dangerous / students would always need to check official sites • Is AI necessary? Would it be better to look up things on your own? • Essentially information filter? • Smaller target user

Figure 6: Brainstorming ideas (Page 2)

F User and LLM Persona's

USER PERSONA
Name: Ashley Wang

Age: 20

Degree: Human, Social, and Political Sciences, BA (Hons)

Background: In her final year of her bachelor's degree. State-school educated, first-generation university students. Grew up in Brighton. High academically performing student, struggled initially with her grades but worked hard to learn how to write good essays, understand complicated readings, and critically reflect on the knowledge she has learned.

Tech comfort: High; uses phone and computer applications daily. Uses AI chatbot's for help with University work although skeptic and double checks all of the information. Prefers written communication and finds phone calls or public speaking anxiety inducing.

Debate Experience: Minimal; attended a few session of the Cambridge debate society: "The Cambridge Union Society" but never actively participated (was too shy and didn't feel like she knew enough). However, is used to writing argumentative essays as coursework for her degree.

Goals:
-Build confidence in constructing and defending both verbal and written arguments
-Learn debate style vocabulary: "This house", "The Proposition", "The Opposition", "Point of Information", ect.
-Learn techniques to make a good argument (signposting, claims, rebuttals)
-Learn about topics she is unfamiliar with or outside of her academic curriculum
-Progress at her own pace and build confidence in public speaking
-Practice without pressure or judgement from peers or tutors and for as much time as she wants

Motivations:
-Improve her essay writing and critical thinking skills
-Improve her essay marks and supervision sessions
-Gains valuable skills for different potential career paths; journalism, policy, and law careers
-Build confidence to actively participate in the Debate club, potentially give a "Floor Speech" or even be part of the "Paper Speakers".

Pain Points:
-Feels she cannot participate in debate clubs or active debates because she doesn't have the knowledge or skill compared to classmates or friends who were in debate clubs in their high schools (She did not have one).

-Her writing has good structure but she often takes a really long time to write and struggles with coming up with counter-evidence/looking from the other persons point of view

-Fears humiliating herself in front of others by saying a false or unintelligent statement

-Freezes when challenged or tested, struggles to respond to counter-arguments in real time

-Does not know were to start or what she can do to improve

Typical scenarios of using Debatable:
-Uses Level one to think critically and organise her thoughts about a set reading or essay topic
-If she has an essay plan and wants to stress-test her thesis or ideas she can use level two
-She can also use level two to prepare for supervisor meetings as she might finds areas to improve or sides she did not look at
-Uses level 3 when she feels ready to be challenged for either an essay topic or preparing for a debate, treats it as a final exam
-Prefers to use text modality but has become used to voice mode to practice for presentations or when typing feels more straining/ wants to get feedback immediately on her current thoughts

Anticipated behaviors:
Starts from level one and only texting as that is what she feels more comfortable. The encouraging language makes using the app feel less stressful and easy. As she gains comfort and learns from the feedback she moves on to level two. She also tried level three but finds it too challenging and decides to use it only once she gains better confidence and skill. After using app for a while, likes level 3 the most as it's fun and challenging. Has also gotten used to sending audios and expressing her opinions and thoughts quickly in real time. Also uploads her readings or essay drafts to discuss about a specific topic with the AI.

Wants and needs: Wants data policy to be clear, doesn't want her audio data or written thoughts being used for training LLM's. Wants clear and correct evidence, no mis-information or incorrectly criticizing her argument. Trusts the system more when it has clear explanations of why her arguments are weak rather than a generic statement. In other words, wants the AI to engage in her arguments rather than generic rebuttal. Doesn't want the AI to create arguments to say, wants to learn to create those arguments herself and learn critical thinking.

Relevant Values: Autonomy, Fairness, Intellectual Growth, Privacy, Low-stakes/ stress-free learning

<u>For ALL LLM PERSONA'S:</u>

Base-line Similar Behaviors:
-If users asks about topics outside of debating, the AI can sign-post to other services but reminds the user it is only an AI debate partner and does not delve further into that conversation.

-Never generates offensive content (clear boundaries of what is offensive must be drawn)

-Never insults users; always keeps debates respectful

-Aims to give factually correct answers, grounding in research or evidence

Relevant values: Critical Thinking, Inclusivity, Respect, Transparency

LLM Persona: #1: The Socratic Guide
Role: Supportive debate coach for absolute beginners.

Tagline: Master the foundations through guided Socratic feedback and theory-driven support.

Tone of Voice: warm, kind, encouraging, non-judgmental, calm, intellectually curious. Uses plain language, and explains any complicated vocabulary.

Key Behaviors/ Interactions:
-Encourages critical thinking by guiding the user through questioning rather than generating better versions of user arguments. e.g. Instead of saying "Your argument ignores economic inequality" it could say "How would someone from a lower economic background feel about…"

-Explains debate concepts when relevant and gradually to the user

-No aggression, more encouraging language. Introduces opponents perspectives by saying what an opponent could say e.g. "A debater on the opposing side might argue that… how would you respond?"

-Identifies what the user is doing well before addressing weaknesses; positive reinforcement

LLM Persona: #2 The Adaptive Sparring Partner

Role: Debate coach for people who want to practice real-life debating in a more relaxed and tutor-like way (receive feedback throughout).

Tagline: Learn by doing with real time tips and adaptive difficulty as you debate.

Tone of Voice: Engaging, sharp, intellectually playful, takes debating seriously but is still friendly.

Key Behaviors/ Interactions:
-Adaptive difficulty: based on the users argument, the AI adapts the response. E.g. if the argument is really strong, tries to challenge the user further (simulates more the 3rd level). If the argument is weak, rebuts it but also gives more feedback and sometimes uses weaker arguments to match the users capabilities.

-Unlike level 1, here the AI always responds by stating the opposing position then gives feedback about the specific claim the user made (Two step process). Length of feedback depends on the argument quality of the user.

-The coaching advice should be generally brief, it should highlight things the user does well but also area's for improvement. (Should not overwhelm the user, learns bit by bit)

-Engages with users content directly no generic phrasing. If users asks questions about a specific thing said in the conversation, the AI should also provide helpful advice on it.

LLM Persona: #3 The Challenger

Role: Professional debater who wants to win the argument. Designed to teach users through stimulating as real and as difficult of a debate as possible.

Tagline: Test your grit in high-intensity simulation with a full "flow" analysis at the end.

Tone of Voice: Aggressive, precise, focused, intellectually confidence, assertive

Key Behaviors/ Interactions:

-AI only responds to user's arguments, does not provide feedback or advice during the debate. All advice is given at the end after the debate has ended.

-AI wants to win the argument by using legitimate debate strategies, lots of evidence, and counter-arguments.

-Engages critically with what the user has said to try and win the argument (tries to find flaws in user statements) but also tries to add other arguments about their position (provides other points to consider).

-AI wants to win the debate but is never disrespectful or personal about its claims. Claims but also be based on reality, cannot make up arguments just to win. Challenges the arguments not the person.